1 (Embedded) Databases Workshop

For the assignment during the databases workshop , you can choose between one assignment within the S6 project or a given assignment as described below (section 2).

1.1 Assignment within the S6 project

When it comes to collecting and storing data in a database within your S6 project, you can use this as part of your Databases workshop assignment. You can then apply the assessment in paragraph 2.4 to your project.

1.2 Lab (Embedded) Databases

For the Lab you can use for example the “Raspberry Pi” as a hardware platform. Other embedded platforms are also allowed.

You can use all the knowledge you have gained in the course IOT in period 3 to retrieve data via LoRaWAN and “The Things Network”. There are various nodes here at school where you can extract data from, such as:

- "Pax counter"
- Temperature
- Light intensity
- Humidity
- Location info (GPS-data)
- Time
- Gateway information
- etc.

The assignments can be done with groups of 2 or 3 people.

1.3 Purpose of the assignments

During this course the student is working on the professional task “Software design and testing” at level 2 (advanced).

The student learns:

- Recognize common database systems and data models and in particular the relational database model (structural, integrity, requirements and behavioural aspects).
- Creating a conceptual designing of a simple relational database by using ER diagrams.
- Creating a logical design and apply the normalization rules.
- Creating an implementation by formulate needs in simple SQL queries.
- Make use of an embedded DBMS, SQLite, in a C or C++ application to build an database.

2 The Assignments

2.1 Task 1

View and study what “SQLite” is and what you can do with it. Install and configure it on the Raspberry Pi Linux platform. Try and experiment with SQLite by using the command line.

2.2 Task 2

Consider what kind of database you want to create on your embedded system. Make a conceptual database design by drawing an ER-diagram. The database must meet at least the following requirements:

- Minimal 3 entities, with a number of attributes per entity.
- Minimal 2 relations

2.3 Task 3

Create a logical database design, before you implement the database, with use of SQLite, into an C or C++ application. Make sure your logical design is in the 3NF. Make also an application that can show, in a well-arranged way, the stored data in the database.

2.4 Assessment

The following items must be delivered and will be assessed in the last week of period 4.

- Conceptual database design (ER-diagrams)
- Logical database design. (Tables, with attributes and keys all in the 3NF).
- Source of an implemented database on the Raspberry Pi with the use of SQLite or other hard/software resources that are used in your project.
- Demonstration of the well-arranged user interface.

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